

Nirvaan Shah

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EDUCATION

University of Toronto

2024 – Expected 2029

B.A.Sc. Civil Engineering with PEY, Engineering Business and Environmental Engineering Minors

Relevant Courses: Structural Analysis, Steel and Timber Design, Reinforced Concrete, Water and Wastewater Treatment Processes, Municipal Engineering, Sustainable Energy Systems, Transport, Engineering Economics, Fundamentals of Accounting and Finance

WORK EXPERIENCE

Engineering Project Analyst Intern *Evardi | Remote, Part-Time Alongside Term*

August 2026– Present

- Building digital models of 3 real power-grid systems, mapping out how substations, power lines, transformers, and renewable energy sources connect to each other. The models are going to simulate what happens during power outages or equipment failures, and to test how reliable each grid is under stress.
- Developing optimization and machine-learning models in Python to analyze grid operating data, identify anomalies, and evaluate control strategies across 50,000+ system measurements and scenarios.

Sustainability and Supply Chain Intern *Testo | Lenzkirch, Germany*

June – August 2026

- Evaluated platinum filtration optimization across 12 work areas and conducted a cost analysis of 5 compliance options spanning €12,000–€30,000, identifying a 15% recovery improvement, recommending a viable approach that met requirements at 70% of the highest-cost alternative and flagging a major environmental concern regarding platinum waste disposal.
- Led a team of 4 interns to audit 30 hazardous material and gas work areas covering 220 regulated substances and 70 machines against government environmental requirements, identifying 45 compliance gaps ahead of a regulatory government inspection.

Aerostructural Design Team Member *UTSM | Toronto, ON, Canada*

September 2024– Present

- Designed and fabricated 12 structural components of a fuel-efficient vehicle body for the Shell Eco-Marathon, an international efficiency competition.
- Modeled components in SOLIDWORKS and fabricated them with laser cutting, 3D printing and power tools, working within a six-person team.

Engineering Consulting Intern *Samir Consulting | Nairobi, Kenya*

June – August 2025

- Supported a six engineering team on the design of a golf course resort development across a 110 hectare site.
- Produced four renderings and three designs using Autocad, Fusion 360, and Revit.

PROJECTS and VOLUNTEERING

Founding Member and Advisor *Still I Rise | Nairobi, Kenya*

March 2023–Present

- One of 15 founding members that created the first free IB school in Mathare slums, Nairobi over 18 months. Now, an advisor supporting outreach for sponsors and fundraising.
- Tutored 60 students across four subjects, mathematics, physics, chemistry and Python, and helped secure \$50,000 in funding and sponsorships.

Engineering Design Mentor *The Tech Interactive | Nairobi, Kenya*

August 2025

- Supported the Drop & Dash Tech Challenge, a hands-on engineering event for more than 10,000 students across South and East Africa.
- Mentored, 80 teams with 4 students each over the span of the week in their challenge of building battery free devices from locally sourced materials, simulating medical supply delivery in remote areas.

Project Manager *ESP Engineering Team | Toronto, ON, Canada*

September 2024– May 2025

- Led a 6 person team designing a solar microgrid for the University of Toronto's sustainability office. Modelled to generate 1.46 GWh annually and save 1,171 tonnes of CO₂, reaching carbon net zero by 2038.
- Modelled energy yield across panel efficiency, tilt, azimuth and various system loss factors to an 81.23 performance ratio%

Project Leader *Plastiki Rafiki | Nairobi, Kenya*

September 2022– May 2024

- Led a student run NGO consisting of 22 members creating and selling items made from recycled plastics, generating \$6500 quarterly, and diverting an estimated 200 Kg of plastic waste a month through community collection programs.
- Personally established four production workshops each employing 20 local residents and providing a wage of \$11 an hour.

TECHNICAL SKILLS

Design & Modelling | AutoCAD, Revit, SOLIDWORKS, Rhino, Fusion 360, SketchUp, Blender, Adobe Illustrator, Adobe Photoshop

Coding and Analysis | Python, MATLAB, Microsoft Excel (VBA, Power Query), ETABS, ArcGIS, Surveying Instrumentation

Physical Skills | Laser Cutting, 3D Printing, Resin Molding, CNC Machining, Plastic Shredding & Extrusion, Power Tools

Certifications and Languages Fluent in English and Swahili, Proficient in Spanish, Hindi, and Gujarati, Red Cross First Aid Certified, International Drivers License